JANNAH JOY ABELLANOSA CPSC 4800 Assignment 2 Part 2

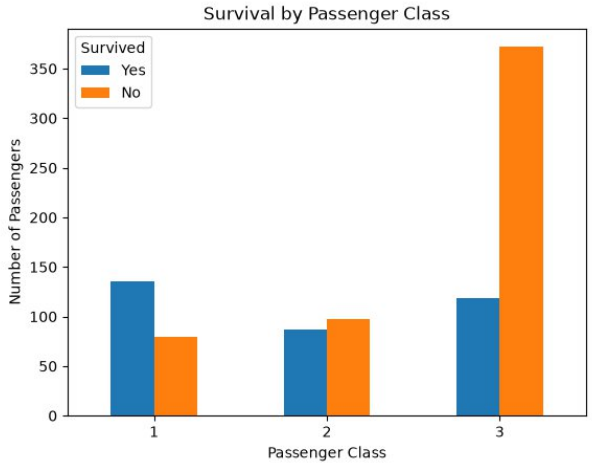
Titanic Survival Analysis

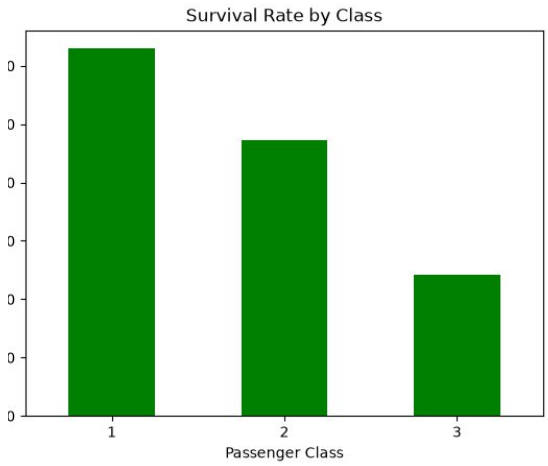
Objective

The purpose of this analysis was to explore factors associated with passenger survival aboard the Titanic. Using exploratory data analysis and visualizations, three hypotheses were evaluated to determine whether survival was associated with passenger class, gender, and age.

Hypothesis 1: Survival Rate and Passenger Class

Finding: Passenger class appears to have a significant relationship with survival. First-class passengers generally had higher survival rates compared to second- and third-class passengers.



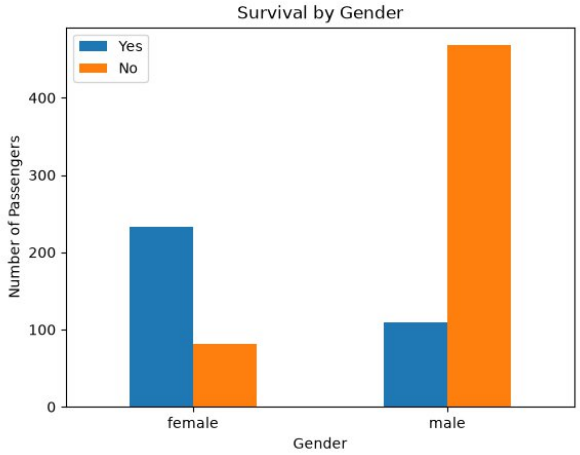


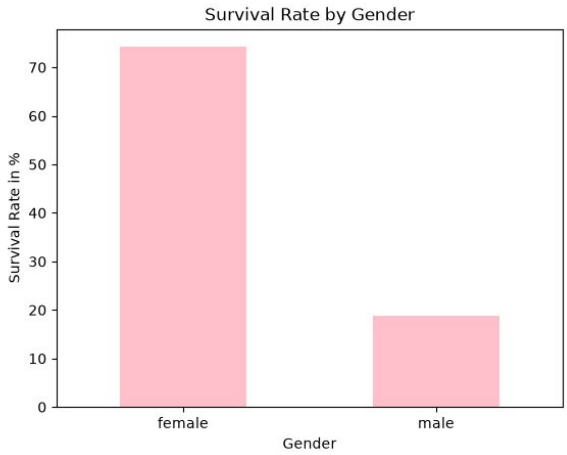
Interpretation:

Higher-class passengers may have had better access to lifeboats and evacuation procedures, leading to increased survival rates.

Hypothesis 2: Survival Rate and Gender

Finding: Female passengers had a substantially higher survival rate compared to male passengers.



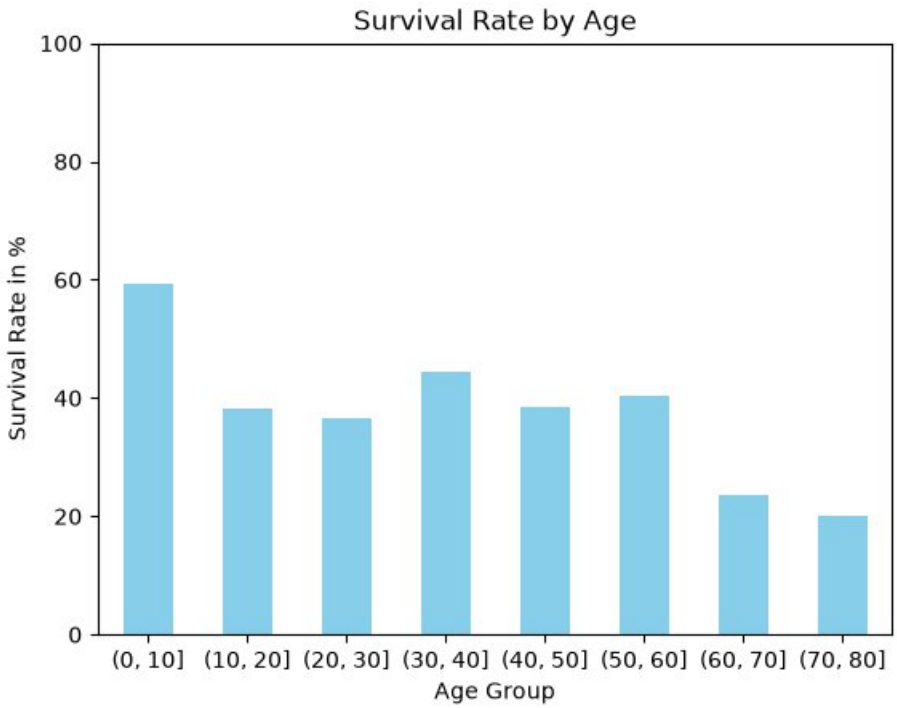


Interpretation:

The results support the historical "women and children first" evacuation practice implemented during the disaster.

Hypothesis 3: Survival Rate and Age

Finding: Survival varied across age groups. Younger passengers generally experienced higher survival rates than older passengers.



Conclusion

The analysis suggests that passenger class, gender, and age were all associated with survival outcomes on the Titanic. Among the factors examined, gender demonstrated the strongest difference in survival rates, followed by passenger class. More women survived more than men, and more passengers from Class 1 survived. These findings highlight how demographic and socioeconomic factors influenced survival during the disaster.